In [1]:
```python
import pandas as pd
```

In [32]:
```python
from tqdm import tqdm
```

In [50]:
```python
import matplotlib.pyplot as plt
```

In [75]:
```python
import numpy as np
```

In [3]:
```python
Data = pd.read_csv('Data/FeatureEngineered_Data.csv')
```

In [4]:
```python
Data.head(20)
```

Out[4]:

	provinceid	dguid	datestamp	noc_groupingid	employment	year	Key	datestampe	ER_t_1	ER_t_2	...	quarter	week	dayofyear_sin	dayofyear_cos	month_sin	month_cos	quarter_sin	quarter_cos	week_sin	week_cos
0	10	2021A000210	2000-01-01	1	7.0	2000	(10, 1)	2000-01-01	NaN	NaN	...	1	52	0.017213	0.999852	5.000000e-01	8.660254e-01	1.000000e+00	6.123234e-17	6.432491e-16	1.000000e+00
1	10	2021A000210	2000-02-01	1	8.2	2000	(10, 1)	2000-02-01	7.0	NaN	...	1	5	0.523416	0.852078	8.660254e-01	5.000000e-01	1.000000e+00	6.123234e-17	5.680647e-01	8.229839e-01
2	10	2021A000210	2000-03-01	1	7.1	2000	(10, 1)	2000-03-01	8.2	7.0	...	1	9	0.867456	0.497513	1.000000e+00	6.123234e-17	1.000000e+00	6.123234e-17	8.854560e-01	4.647232e-01
3	10	2021A000210	2000-04-01	1	7.6	2000	(10, 1)	2000-04-01	7.1	8.2	...	2	13	0.999917	-0.012910	8.660254e-01	-5.000000e-01	1.224647e-16	-1.000000e+00	1.000000e+00	-1.608123e-16
4	10	2021A000210	2000-05-01	1	6.9	2000	(10, 1)	2000-05-01	7.6	7.1	...	2	18	0.863142	-0.504961	5.000000e-01	-8.660254e-01	1.224647e-16	-1.000000e+00	8.229839e-01	-5.680647e-01
5	10	2021A000210	2000-06-01	1	7.8	2000	(10, 1)	2000-06-01	6.9	7.6	...	2	22	0.486273	-0.873807	1.224647e-16	-1.000000e+00	1.224647e-16	-1.000000e+00	4.647232e-01	-8.854560e-01
6	10	2021A000210	2000-07-01	1	7.3	2000	(10, 1)	2000-07-01	7.8	6.9	...	3	26	-0.008607	-0.999963	-5.000000e-01	-8.660254e-01	-1.000000e+00	-1.836970e-16	-3.216245e-16	-1.000000e+00
7	10	2021A000210	2000-08-01	1	7.0	2000	(10, 1)	2000-08-01	7.3	7.8	...	3	31	-0.516062	-0.856551	-8.660254e-01	-5.000000e-01	-1.000000e+00	-1.836970e-16	-5.680647e-01	-8.229839e-01
8	10	2021A000210	2000-09-01	1	7.9	2000	(10, 1)	2000-09-01	7.0	7.3	...	3	35	-0.880012	-0.474951	-1.000000e+00	-1.836970e-16	-1.000000e+00	-1.836970e-16	-8.854560e-01	-4.647232e-01
9	10	2021A000210	2000-10-01	1	8.4	2000	(10, 1)	2000-10-01	7.9	7.0	...	4	39	-0.999769	0.021516	-8.660254e-01	5.000000e-01	-2.449294e-16	1.000000e+00	-1.000000e+00	-1.836970e-16
10	10	2021A000210	2000-11-01	1	7.8	2000	(10, 1)	2000-11-01	8.4	7.9	...	4	44	-0.849817	0.527078	-5.000000e-01	8.660254e-01	-2.449294e-16	1.000000e+00	-8.229839e-01	5.680647e-01
11	10	2021A000210	2000-12-01	1	8.0	2000	(10, 1)	2000-12-01	7.8	8.4	...	4	48	-0.478734	0.877960	-2.449294e-16	1.000000e+00	-2.449294e-16	1.000000e+00	-4.647232e-01	8.854560e-01
12	10	2021A000210	2001-01-01	1	8.2	2001	(10, 1)	2001-01-01	8.0	7.8	...	1	1	0.017213	0.999852	5.000000e-01	8.660254e-01	1.000000e+00	6.123234e-17	1.205367e-01	9.927089e-01
13	10	2021A000210	2001-02-01	1	7.6	2001	(10, 1)	2001-02-01	8.2	8.0	...	1	5	0.523416	0.852078	8.660254e-01	5.000000e-01	1.000000e+00	6.123234e-17	5.680647e-01	8.229839e-01
14	10	2021A000210	2001-03-01	1	6.7	2001	(10, 1)	2001-03-01	7.6	8.2	...	1	9	0.858764	0.512371	1.000000e+00	6.123234e-17	1.000000e+00	6.123234e-17	8.854560e-01	4.647232e-01
15	10	2021A000210	2001-04-01	1	7.1	2001	(10, 1)	2001-04-01	6.7	7.6	...	2	13	0.999991	0.004304	8.660254e-01	-5.000000e-01	1.224647e-16	-1.000000e+00	1.000000e+00	-1.608123e-16
16	10	2021A000210	2001-05-01	1	7.4	2001	(10, 1)	2001-05-01	7.1	6.7	...	2	18	0.871706	-0.490029	5.000000e-01	-8.660254e-01	1.224647e-16	-1.000000e+00	8.229839e-01	-5.680647e-01
17	10	2021A000210	2001-06-01	1	7.2	2001	(10, 1)	2001-06-01	7.4	7.1	...	2	22	0.501242	-0.865307	1.224647e-16	-1.000000e+00	1.224647e-16	-1.000000e+00	4.647232e-01	-8.854560e-01
18	10	2021A000210	2001-07-01	1	7.0	2001	(10, 1)	2001-07-01	7.2	7.4	...	3	26	0.008607	-0.999963	-5.000000e-01	-8.660254e-01	-1.000000e+00	-1.836970e-16	-3.216245e-16	-1.000000e+00
19	10	2021A000210	2001-08-01	1	6.7	2001	(10, 1)	2001-08-01	7.0	7.2	...	3	31	-0.501242	-0.865307	-8.660254e-01	-5.000000e-01	-1.000000e+00	-1.836970e-16	-5.680647e-01	-8.229839e-01

20 rows × 56 columns

In [5]:
```python
print(f"Data Columns =>\n{list(Data.columns)}")
```

```
Data Columns =>
['provinceid', 'dguid', 'datestamp', 'noc_groupingid', 'employment', 'year', 'Key', 'datestampe', 'ER_t_1', 'ER_t_2', 'ER_t_3', 'ER_t_4', 'ER_t_5', 'ER_t_6', 'ER_t_7', 'ER_t_8', 'ER_t_9', 'ER_t_10', 'ER_t_11', 'ER_t_12', 'ER_t_1_rollingmean', 'ER_t_1_rollingstd', 'ER_t_2_rollingmean', 'ER_t_2_rollingstd', 'ER_t_3_rollingmean', 'ER_t_3_rollingstd', 'ER_t_4_rollingmean', 'ER_t_4_rollingstd', 'ER_t_5_rollingme
an', 'ER_t_5_rollingstd', 'ER_t_6_rollingmean', 'ER_t_6_rollingstd', 'ER_t_7_rollingmean', 'ER_t_7_rollingstd', 'ER_t_8_rollingmean', 'ER_t_8_rollingstd', 'ER_t_9_rollingmean', 'ER_t_9_rollingstd', 'ER_t_10_rollingmean', 'ER_t_10_rollingstd', 'ER_t_11_rollingmean', 'ER_t_11_rollingstd', 'ER_t_12_rollingmean', 'ER_t_12_rollingstd', 'dayofyear', 'month', 'quarter', 'week', 'dayofyear_sin', 'dayofyear_cos', 'm
onth_sin', 'month_cos', 'quarter_sin', 'quarter_cos', 'week_sin', 'week_cos']
```

In [6]:
```python
Data.columns.tolist()
```

Out[6]:
```
['provinceid',
 'dguid',
 'datestamp',
 'noc_groupingid',
 'employment',
 'year',
 'Key',
 'datestampe',
 'ER_t_1',
 'ER_t_2',
 'ER_t_3',
 'ER_t_4',
 'ER_t_5',
 'ER_t_6',
 'ER_t_7',
 'ER_t_8',
 'ER_t_9',
 'ER_t_10',
 'ER_t_11',
 'ER_t_12',
 'ER_t_1_rollingmean',
 'ER_t_1_rollingstd',
 'ER_t_2_rollingmean',
 'ER_t_2_rollingstd',
 'ER_t_3_rollingmean',
 'ER_t_3_rollingstd',
 'ER_t_4_rollingmean',
 'ER_t_4_rollingstd',
 'ER_t_5_rollingmean',
 'ER_t_5_rollingstd',
 'ER_t_6_rollingmean',
 'ER_t_6_rollingstd',
 'ER_t_7_rollingmean',
 'ER_t_7_rollingstd',
 'ER_t_8_rollingmean',
 'ER_t_8_rollingstd',
 'ER_t_9_rollingmean',
 'ER_t_9_rollingstd',
 'ER_t_10_rollingmean',
 'ER_t_10_rollingstd',
 'ER_t_11_rollingmean',
 'ER_t_11_rollingstd',
 'ER_t_12_rollingmean',
 'ER_t_12_rollingstd',
 'dayofyear',
 'month',
 'quarter',
 'week',
 'dayofyear_sin',
 'dayofyear_cos',
 'month_sin',
 'month_cos',
 'quarter_sin',
 'quarter_cos',
 'week_sin',
 'week_cos']
```

In [7]:
```python
lags_and_indicators = [ 'ER_t_1',
 'ER_t_2',
 'ER_t_3',
 'ER_t_4',
 'ER_t_5',
 'ER_t_6',
 'ER_t_7',
 'ER_t_8',
 'ER_t_9',
 'ER_t_10',
 'ER_t_11',
 'ER_t_12',
 'ER_t_1_rollingmean',
 'ER_t_1_rollingstd',
 'ER_t_2_rollingmean',
 'ER_t_2_rollingstd',
 'ER_t_3_rollingmean',
 'ER_t_3_rollingstd',
 'ER_t_4_rollingmean',
 'ER_t_4_rollingstd',
 'ER_t_5_rollingmean',
 'ER_t_5_rollingstd',
 'ER_t_6_rollingmean',
 'ER_t_6_rollingstd',
 'ER_t_7_rollingmean',
 'ER_t_7_rollingstd',
 'ER_t_8_rollingmean',
 'ER_t_8_rollingstd',
 'ER_t_9_rollingmean',
 'ER_t_9_rollingstd',
 'ER_t_10_rollingmean',
 'ER_t_10_rollingstd',
 'ER_t_11_rollingmean',
 'ER_t_11_rollingstd',
 'ER_t_12_rollingmean',
 'ER_t_12_rollingstd']

exog_variables = [
 'dayofyear_sin',
 'dayofyear_cos',
 'month_sin',
 'month_cos',
 'quarter_sin',
 'quarter_cos',
 'week_sin',
 'week_cos'
]

categoricals = [
 'provinceid',
 'dayofyear',
 'month',
 'quarter',
 'week'
]

target = ['employment']
```

In [8]:
```python
null_counts = Data.groupby('year')[lags_and_indicators].apply(lambda x: x.isna().sum())
null_counts
```

Out[8]:

year	ER_t_1	ER_t_2	ER_t_3	ER_t_4	ER_t_5	ER_t_6	ER_t_7	ER_t_8	ER_t_9	ER_t_10	...	ER_t_8_rollingmean	ER_t_8_rollingstd	ER_t_9_rollingmean	ER_t_9_rollingstd	ER_t_10_rollingmean	ER_t_10_rollingstd	ER_t_11_rollingmean	ER_t_11_rollingstd	ER_t_12_rollingmean	ER_t_12_rollingstd
2000	574	1148	1722	2296	2870	3444	4018	4592	5166	5740	...	6313	6313	6886	6886	6886	6886	6886	6886	6886	6886
2001	0	0	0	0	0	0	0	0	0	0	...	1	1	2	2	576	576	1150	1150	1724	1724
2002	0	0	0	0	0	0	0	0	0	0	...	0	0	0	0	0	0	0	0	0	0
2003	0	0	0	0	0	0	0	0	0	0	...	0	0	0	0	0	0	0	0	0	0
2004	0	0	0	0	0	0	0	0	0	0	...	0	0	0	0	0	0	0	0	0	0
2005	0	0	0	0	0	0	0	0	0	0	...	0	0	0	0	0	0	0	0	0	0
2006	0	0	0	0	0	0	0	0	0	0	...	0	0	0	0	0	0	0	0	0	0
2007	0	0	0	0	0	0	0	0	0	0	...	0	0	0	0	0	0	0	0	0	0
2008	0	0	0	0	0	0	0	0	0	0	...	0	0	0	0	0	0	0	0	0	0
2009	0	0	0	0	0	0	0	0	0	0	...	0	0	0	0	0	0	0	0	0	0
2010	0	0	0	0	0	0	0	0	0	0	...	0	0	0	0	0	0	0	0	0	0
2011	0	0	0	0	0	0	0	0	0	0	...	0	0	0	0	0	0	0	0	0	0
2012	0	0	0	0	0	0	0	0	0	0	...	0	0	0	0	0	0	0	0	0	0
2013	0	0	0	0	0	0	0	0	0	0	...	0	0	0	0	0	0	0	0	0	0
2014	0	0	0	0	0	0	0	0	0	0	...	0	0	0	0	0	0	0	0	0	0
2015	0	0	0	0	0	0	0	0	0	0	...	0	0	0	0	0	0	0	0	0	0
2016	0	0	0	0	0	0	0	0	0	0	...	0	0	0	0	0	0	0	0	0	0
2017	0	0	0	0	0	0	0	0	0	0	...	0	0	0	0	0	0	0	0	0	0
2018	0	0	0	0	0	0	0	0	0	0	...	0	0	0	0	0	0	0	0	0	0
2019	0	0	0	0	0	0	0	0	0	0	...	0	0	0	0	0	0	0	0	0	0
2020	0	0	0	0	0	0	0	0	0	0	...	0	0	0	0	0	0	0	0	0	0
2021	0	0	0	0	0	0	0	0	0	0	...	0	0	0	0	0	0	0	0	0	0
2022	0	0	0	0	0	0	0	0	0	0	...	0	0	0	0	0	0	0	0	0	0
2023	0	0	0	0	0	0	0	0	0	0	...	0	0	0	0	0	0	0	0	0	0
2024	0	0	0	0	0	0	0	0	0	0	...	0	0	0	0	0	0	0	0	0	0
2025	0	0	0	0	0	0	0	0	0	0	...	0	0	0	0	0	0	0	0	0	0

26 rows × 36 columns

In [13]:
```python
#Training should start at 2002
```

In [11]:
```python
train = Data[(Data['year']>=2002) & (Data['year']<2025)] #Includes intented training and validation periods.
train.head(50)
```

Out[11]:

	provinceid	dguid	datestamp	noc_groupingid	employment	year	Key	datestampe	ER_t_1	ER_t_2	...	quarter	week	dayofyear_sin	dayofyear_cos	month_sin	month_cos	quarter_sin	quarter_cos	week_sin	week_cos
24	10	2021A000210	2002-01-01	1	9.5	2002	(10, 1)	2002-01-01	9.0	7.8	...	1	1	0.017213	0.999852	5.000000e-01	8.660254e-01	1.000000e+00	6.123234e-17	1.205367e-01	9.927089e-01
25	10	2021A000210	2002-02-01	1	9.3	2002	(10, 1)	2002-02-01	9.5	9.0	...	1	5	0.523416	0.852078	8.660254e-01	5.000000e-01	1.000000e+00	6.123234e-17	5.680647e-01	8.229839e-01
26	10	2021A000210	2002-03-01	1	8.1	2002	(10, 1)	2002-03-01	9.3	9.5	...	1	9	0.858764	0.512371	1.000000e+00	6.123234e-17	1.000000e+00	6.123234e-17	8.854560e-01	4.647232e-01
27	10	2021A000210	2002-04-01	1	8.3	2002	(10, 1)	2002-04-01	8.1	9.3	...	2	14	0.999991	0.004304	8.660254e-01	-5.000000e-01	1.224647e-16	-1.000000e+00	9.927089e-01	-1.205367e-01
28	10	2021A000210	2002-05-01	1	9.0	2002	(10, 1)	2002-05-01	8.3	8.1	...	2	18	0.871706	-0.490029	5.000000e-01	-8.660254e-01	1.224647e-16	-1.000000e+00	8.229839e-01	-5.680647e-01
29	10	2021A000210	2002-06-01	1	8.7	2002	(10, 1)	2002-06-01	9.0	8.3	...	2	22	0.501242	-0.865307	1.224647e-16	-1.000000e+00	1.224647e-16	-1.000000e+00	4.647232e-01	-8.854560e-01
30	10	2021A000210	2002-07-01	1	8.2	2002	(10, 1)	2002-07-01	8.7	9.0	...	3	27	0.008607	-0.999963	-5.000000e-01	-8.660254e-01	-1.000000e+00	-1.836970e-16	-1.205367e-01	-9.927089e-01
31	10	2021A000210	2002-08-01	1	8.9	2002	(10, 1)	2002-08-01	8.2	8.7	...	3	31	-0.501242	-0.865307	-8.660254e-01	-5.000000e-01	-1.000000e+00	-1.836970e-16	-5.680647e-01	-8.229839e-01
32	10	2021A000210	2002-09-01	1	9.0	2002	(10, 1)	2002-09-01	8.9	8.2	...	3	35	-0.871706	-0.490029	-1.000000e+00	-1.836970e-16	-1.000000e+00	-1.836970e-16	-8.854560e-01	-4.647232e-01
33	10	2021A000210	2002-10-01	1	8.2	2002	(10, 1)	2002-10-01	9.0	8.9	...	4	40	-0.999991	0.004304	-8.660254e-01	5.000000e-01	-2.449294e-16	1.000000e+00	-9.927089e-01	1.205367e-01
34	10	2021A000210	2002-11-01	1	7.6	2002	(10, 1)	2002-11-01	8.2	9.0	...	4	44	-0.858764	0.512371	-5.000000e-01	8.660254e-01	-2.449294e-16	1.000000e+00	-8.229839e-01	5.680647e-01
35	10	2021A000210	2002-12-01	1	8.3	2002	(10, 1)	2002-12-01	7.6	8.2	...	4	48	-0.493776	0.869589	-2.449294e-16	1.000000e+00	-2.449294e-16	1.000000e+00	-4.647232e-01	8.854560e-01
36	10	2021A000210	2003-01-01	1	8.9	2003	(10, 1)	2003-01-01	8.3	7.6	...	1	1	0.017213	0.999852	5.000000e-01	8.660254e-01	1.000000e+00	6.123234e-17	1.205367e-01	9.927089e-01
37	10	2021A000210	2003-02-01	1	8.5	2003	(10, 1)	2003-02-01	8.9	8.3	...	1	5	0.523416	0.852078	8.660254e-01	5.000000e-01	1.000000e+00	6.123234e-17	5.680647e-01	8.229839e-01
38	10	2021A000210	2003-03-01	1	7.9	2003	(10, 1)	2003-03-01	8.5	8.9	...	1	9	0.858764	0.512371	1.000000e+00	6.123234e-17	1.000000e+00	6.123234e-17	8.854560e-01	4.647232e-01
39	10	2021A000210	2003-04-01	1	8.7	2003	(10, 1)	2003-04-01	7.9	8.5	...	2	14	0.999991	0.004304	8.660254e-01	-5.000000e-01	1.224647e-16	-1.000000e+00	9.927089e-01	-1.205367e-01
40	10	2021A000210	2003-05-01	1	9.0	2003	(10, 1)	2003-05-01	8.7	7.9	...	2	18	0.871706	-0.490029	5.000000e-01	-8.660254e-01	1.224647e-16	-1.000000e+00	8.229839e-01	-5.680647e-01
41	10	2021A000210	2003-06-01	1	9.1	2003	(10, 1)	2003-06-01	9.0	8.7	...	2	22	0.501242	-0.865307	1.224647e-16	-1.000000e+00	1.224647e-16	-1.000000e+00	4.647232e-01	-8.854560e-01
42	10	2021A000210	2003-07-01	1	9.4	2003	(10, 1)	2003-07-01	9.1	9.0	...	3	27	0.008607	-0.999963	-5.000000e-01	-8.660254e-01	-1.000000e+00	-1.836970e-16	-1.205367e-01	-9.927089e-01
43	10	2021A000210	2003-08-01	1	9.7	2003	(10, 1)	2003-08-01	9.4	9.1	...	3	31	-0.501242	-0.865307	-8.660254e-01	-5.000000e-01	-1.000000e+00	-1.836970e-16	-5.680647e-01	-8.229839e-01
44	10	2021A000210	2003-09-01	1	9.4	2003	(10, 1)	2003-09-01	9.7	9.4	...	3	36	-0.871706	-0.490029	-1.000000e+00	-1.836970e-16	-1.000000e+00	-1.836970e-16	-9.350162e-01	-3.546049e-01
45	10	2021A000210	2003-10-01	1	7.4	2003	(10, 1)	2003-10-01	9.4	9.7	...	4	40	-0.999991	0.004304	-8.660254e-01	5.000000e-01	-2.449294e-16	1.000000e+00	-9.927089e-01	1.205367e-01
46	10	2021A000210	2003-11-01	1	6.4	2003	(10, 1)	2003-11-01	7.4	9.4	...	4	44	-0.858764	0.512371	-5.000000e-01	8.660254e-01	-2.449294e-16	1.000000e+00	-8.229839e-01	5.680647e-01
47	10	2021A000210	2003-12-01	1	7.0	2003	(10, 1)	2003-12-01	6.4	7.4	...	4	49	-0.493776	0.869589	-2.449294e-16	1.000000e+00	-2.449294e-16	1.000000e+00	-3.546049e-01	9.350162e-01
48	10	2021A000210	2004-01-01	1	6.6	2004	(10, 1)	2004-01-01	7.0	6.4	...	1	1	0.017213	0.999852	5.000000e-01	8.660254e-01	1.000000e+00	6.123234e-17	1.205367e-01	9.927089e-01
49	10	2021A000210	2004-02-01	1	5.9	2004	(10, 1)	2004-02-01	6.6	7.0	...	1	5	0.523416	0.852078	8.660254e-01	5.000000e-01	1.000000e+00	6.123234e-17	5.680647e-01	8.229839e-01
50	10	2021A000210	2004-03-01	1	5.9	2004	(10, 1)	2004-03-01	5.9	6.6	...	1	10	0.867456	0.497513	1.000000e+00	6.123234e-17	1.000000e+00	6.123234e-17	9.350162e-01	3.546049e-01
51	10	2021A000210	2004-04-01	1	6.0	2004	(10, 1)	2004-04-01	5.9	5.9	...	2	14	0.999917	-0.012910	8.660254e-01	-5.000000e-01	1.224647e-16	-1.000000e+00	9.927089e-01	-1.205367e-01
52	10	2021A000210	2004-05-01	1	7.3	2004	(10, 1)	2004-05-01	6.0	5.9	...	2	18	0.863142	-0.504961	5.000000e-01	-8.660254e-01	1.224647e-16	-1.000000e+00	8.229839e-01	-5.680647e-01
53	10	2021A000210	2004-06-01	1	7.2	2004	(10, 1)	2004-06-01	7.3	6.0	...	2	23	0.486273	-0.873807	1.224647e-16	-1.000000e+00	1.224647e-16	-1.000000e+00	3.546049e-01	-9.350162e-01
54	10	2021A000210	2004-07-01	1	7.1	2004	(10, 1)	2004-07-01	7.2	7.3	...	3	27	-0.008607	-0.999963	-5.000000e-01	-8.660254e-01	-1.000000e+00	-1.836970e-16	-1.205367e-01	-9.927089e-01
55	10	2021A000210	2004-08-01	1	7.5	2004	(10, 1)	2004-08-01	7.1	7.2	...	3	31	-0.516062	-0.856551	-8.660254e-01	-5.000000e-01	-1.000000e+00	-1.836970e-16	-5.680647e-01	-8.229839e-01
56	10	2021A000210	2004-09-01	1	6.9	2004	(10, 1)	2004-09-01	7.5	7.1	...	3	36	-0.880012	-0.474951	-1.000000e+00	-1.836970e-16	-1.000000e+00	-1.836970e-16	-9.350162e-01	-3.546049e-01
57	10	2021A000210	2004-10-01	1	7.8	2004	(10, 1)	2004-10-01	6.9	7.5	...	4	40	-0.999769	0.021516	-8.660254e-01	5.000000e-01	-2.449294e-16	1.000000e+00	-9.927089e-01	1.205367e-01
58	10	2021A000210	2004-11-01	1	7.2	2004	(10, 1)	2004-11-01	7.8	6.9	...	4	45	-0.849817	0.527078	-5.000000e-01	8.660254e-01	-2.449294e-16	1.000000e+00	-7.485107e-01	6.631227e-01
59	10	2021A000210	2004-12-01	1	7.7	2004	(10, 1)	2004-12-01	7.2	7.8	...	4	49	-0.478734	0.877960	-2.449294e-16	1.000000e+00	-2.449294e-16	1.000000e+00	-3.546049e-01	9.350162e-01
60	10	2021A000210	2005-01-01	1	8.1	2005	(10, 1)	2005-01-01	7.7	7.2	...	1	53	0.017213	0.999852	5.000000e-01	8.660254e-01	1.000000e+00	6.123234e-17	1.205367e-01	9.927089e-01
61	10	2021A000210	2005-02-01	1	8.0	2005	(10, 1)	2005-02-01	8.1	7.7	...	1	5	0.523416	0.852078	8.660254e-01	5.000000e-01	1.000000e+00	6.123234e-17	5.680647e-01	8.229839e-01
62	10	2021A000210	2005-03-01	1	8.2	2005	(10, 1)	2005-03-01	8.0	8.1	...	1	9	0.858764	0.512371	1.000000e+00	6.123234e-17	1.000000e+00	6.123234e-17	8.854560e-01	4.647232e-01
63	10	2021A000210	2005-04-01	1	7.1	2005	(10, 1)	2005-04-01	8.2	8.0	...	2	13	0.999991	0.004304	8.660254e-01	-5.000000e-01	1.224647e-16	-1.000000e+00	1.000000e+00	-1.608123e-16
64	10	2021A000210	2005-05-01	1	8.0	2005	(10, 1)	2005-05-01	7.1	8.2	...	2	17	0.871706	-0.490029	5.000000e-01	-8.660254e-01	1.224647e-16	-1.000000e+00	8.854560e-01	-4.647232e-01
65	10	2021A000210	2005-06-01	1	8.0	2005	(10, 1)	2005-06-01	8.0	7.1	...	2	22	0.501242	-0.865307	1.224647e-16	-1.000000e+00	1.224647e-16	-1.000000e+00	4.647232e-01	-8.854560e-01
66	10	2021A000210	2005-07-01	1	8.7	2005	(10, 1)	2005-07-01	8.0	8.0	...	3	26	0.008607	-0.999963	-5.000000e-01	-8.660254e-01	-1.000000e+00	-1.836970e-16	-3.216245e-16	-1.000000e+00
67	10	2021A000210	2005-08-01	1	7.6	2005	(10, 1)	2005-08-01	8.7	8.0	...	3	31	-0.501242	-0.865307	-8.660254e-01	-5.000000e-01	-1.000000e+00	-1.836970e-16	-5.680647e-01	-8.229839e-01
68	10	2021A000210	2005-09-01	1	8.8	2005	(10, 1)	2005-09-01	7.6	8.7	...	3	35	-0.871706	-0.490029	-1.000000e+00	-1.836970e-16	-1.000000e+00	-1.836970e-16	-8.854560e-01	-4.647232e-01
69	10	2021A000210	2005-10-01	1	8.1	2005	(10, 1)	2005-10-01	8.8	7.6	...	4	39	-0.999991	0.004304	-8.660254e-01	5.000000e-01	-2.449294e-16	1.000000e+00	-1.000000e+00	-1.836970e-16
70	10	2021A000210	2005-11-01	1	7.9	2005	(10, 1)	2005-11-01	8.1	8.8	...	4	44	-0.858764	0.512371	-5.000000e-01	8.660254e-01	-2.449294e-16	1.000000e+00	-8.229839e-01	5.680647e-01
71	10	2021A000210	2005-12-01	1	7.9	2005	(10, 1)	2005-12-01	7.9	8.1	...	4	48	-0.493776	0.869589	-2.449294e-16	1.000000e+00	-2.449294e-16	1.000000e+00	-4.647232e-01	8.854560e-01
72	10	2021A000210	2006-01-01	1	8.9	2006	(10, 1)	2006-01-01	7.9	7.9	...	1	52	0.017213	0.999852	5.000000e-01	8.660254e-01	1.000000e+00	6.123234e-17	6.432491e-16	1.000000e+00
73	10	2021A000210	2006-02-01	1	9.0	2006	(10, 1)	2006-02-01	8.9	7.9	...	1	5	0.523416	0.852078	8.660254e-01	5.000000e-01	1.000000e+00	6.123234e-17	5.680647e-01	8.229839e-01

50 rows × 56 columns

Filter Methods -- Start -- Testing Correlations

In [42]:
```python
Keys = train['Key'].unique().tolist()
```

In [44]:
```python
def Generate_Correlations(Sample):
    Corrs = Sample.corr()['employment']
    return Corrs

def Append_Corrs(Corrs, Corr_List, Feats, Key):
    corr_dict = {}
    corr_dict['Key'] = Key

    for f in Feats:
        corr_dict[f] = Corrs[f]

    Corr_List.append(corr_dict)

def Store_Correlations(Keys, train):
    Corrs_List = []

    for key in tqdm(Keys, desc='Processing NOC Groups', unit='NOC Grouping ID'):
        Sample = train[train['Key']==key][lags_and_indicators + target]

        Corrs = Generate_Correlations(Sample=Sample)
        Append_Corrs(Corrs=Corrs, Corr_List=Corrs_List, Feats=lags_and_indicators, Key=key)

    df_Corrs = pd.DataFrame(Corrs_List)
    return df_Corrs
```

In [45]:
```python
df_Corrs = Store_Correlations(Keys=Keys, train=train)
```

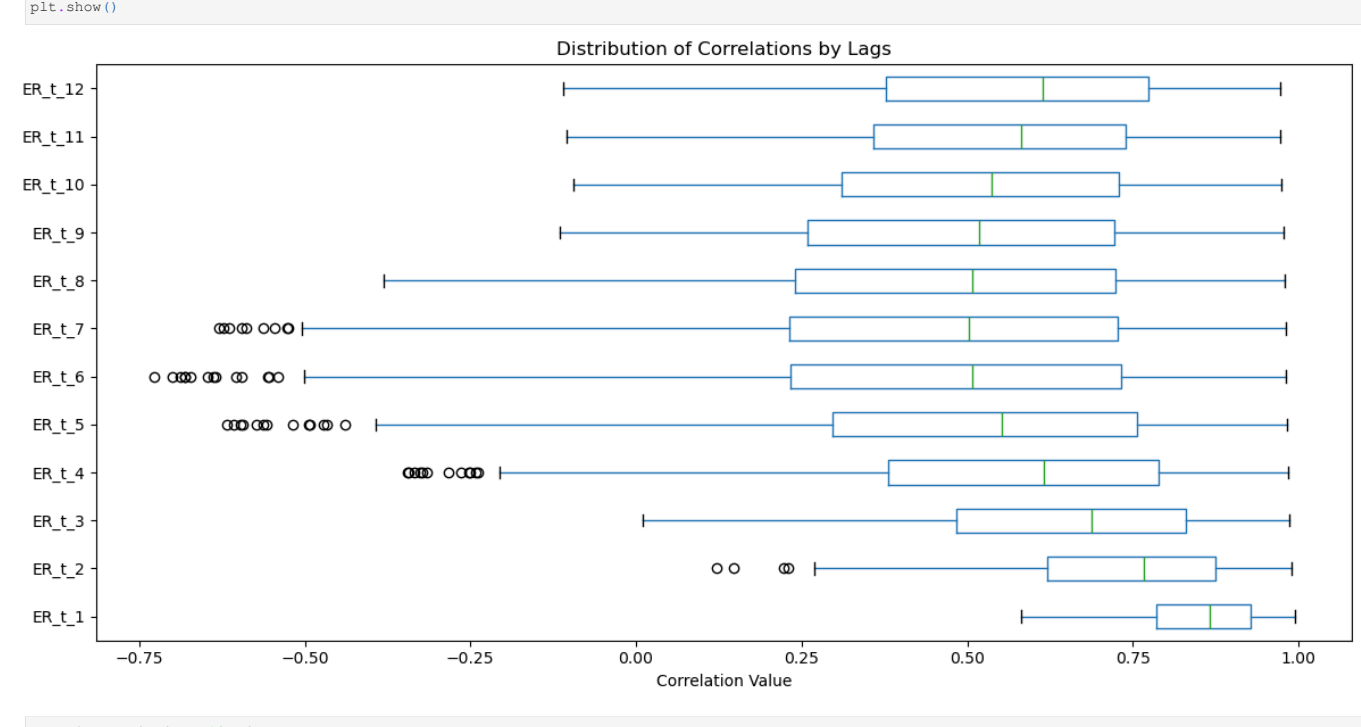
```
Processing NOC Groups: 100%|██████████| 574/574 [00:03<00:00, 182.32NOC Grouping ID/s]
```

In [55]:
```python
lags = ['ER_t_1',
 'ER_t_2',
 'ER_t_3',
 'ER_t_4',
 'ER_t_5',
 'ER_t_6',
 'ER_t_7',
 'ER_t_8',
 'ER_t_9',
 'ER_t_10',
 'ER_t_11',
 'ER_t_12',]
```

In [58]:
```python
indicators = [
 'ER_t_1_rollingmean',
 'ER_t_1_rollingstd',
 'ER_t_2_rollingmean',
 'ER_t_2_rollingstd',
 'ER_t_3_rollingmean',
 'ER_t_3_rollingstd',
 'ER_t_4_rollingmean',
 'ER_t_4_rollingstd',
 'ER_t_5_rollingmean',
 'ER_t_5_rollingstd',
 'ER_t_6_rollingmean',
 'ER_t_6_rollingstd',
 'ER_t_7_rollingmean',
 'ER_t_7_rollingstd',
 'ER_t_8_rollingmean',
 'ER_t_8_rollingstd',
 'ER_t_9_rollingmean',
 'ER_t_9_rollingstd',
 'ER_t_10_rollingmean',
 'ER_t_10_rollingstd',
 'ER_t_11_rollingmean',
 'ER_t_11_rollingstd',
 'ER_t_12_rollingmean',
 'ER_t_12_rollingstd'
]
```

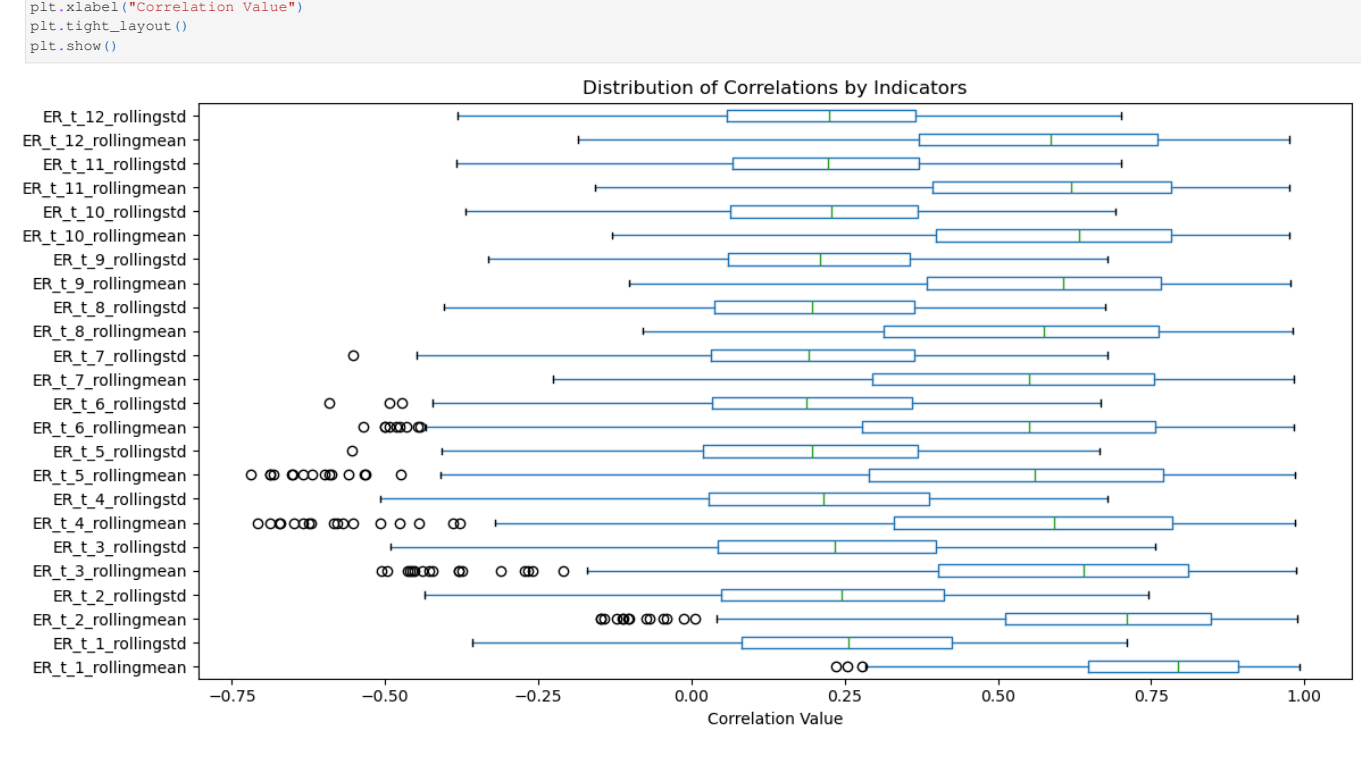
In [57]:
```python
plt.figure(figsize=(12, 6))
df_Corrs[lags].boxplot(
    vert=False,
    grid=False,
    showfliers=True
)

plt.title("Distribution of Correlations by Lags")
plt.xlabel("Correlation Value")
plt.tight_layout()
plt.show()
```



In [59]:
```python
plt.figure(figsize=(12, 6))
df_Corrs[indicators].boxplot(
    vert=False,
    grid=False,
    showfliers=True
)

plt.title("Distribution of Correlations by Indicators")
plt.xlabel("Correlation Value")
plt.tight_layout()
plt.show()
```



Filter Methods -- End -- Outcomes

In [ ]:
```python
'''
Per the box plot distribution it seems that Short Terms lags (1, 2, and 3) along with long term lags (10, 11, 12) provide
best general performance -- in terms of explaining the variance in the target variable.

As such the indicators seem to show that moving averages of the Short Term and Long Terms lags have higher correlations
with the target. Moving standard deviations do not seem to have much to offer.

'''
```

Wrapper Methods -- Start -- Testing Recursive Feature Elimination with XGB

In [63]:
```python
from xgboost import XGBRegressor
from sklearn.feature_selection import RFE
```

In [ ]:
```python
def Initialize_Models():
    #XG Boost Model with Conservative hyper params:
    #   -n_estimators => build 200 trees
    #   -max_depth => model only 4 levels of feature interaction
    #   -learing rate => learn relatively slow
    xgb = XGBRegressor(
    n_estimators=200,
    max_depth=4,
    learning_rate=0.05,
    subsample=0.8,
    colsample_bytree=0.8,
    random_state=42
    )

    #Selecting the best 10 features.
    rfe = RFE(
    estimator=xgb,
    n_features_to_select=10,
    step=1
    )

    return rfe

def Run_Elimination(rfe, X, Y):
    rfe.fit(X, Y)
    return rfe

def Get_Sets(Key):
    train_reduced = train[train['Key']==Key][lags_and_indicators + categoricals + exog_variables + target]
    X = train_reduced[lags_and_indicators + categoricals + exog_variables]
    Y = train_reduced[target]

    return X, Y

def Get_Gain_Importance(rfe, X, Key):
    xgb_importance_full = np.zeros(X.shape[1])

    xgb_importance_full[rfe.support_] = rfe.estimator_.feature_importances_

    summary = pd.DataFrame({
    'Key': Key,
    'feature': X.columns,
    'selected': rfe.support_,
    'rfe_ranking': rfe.ranking_,
    'xgb_gain_importance': xgb_importance_full
    }).sort_values(
    ['selected', 'xgb_gain_importance'],
    ascending=[False, False]
    )

    return summary



def Run_Wrapper_Method(Keys=Keys):

    Feat_Importance_Frame = []
    for Key in tqdm(Keys, desc='Processing Keys of Wrapper Method', unit='Key'):
        X, Y = Get_Sets(Key=Key)

        rfe = Initialize_Models()
        rfe = Run_Elimination(rfe=rfe, X=X, Y=Y)
        summary = Get_Gain_Importance(rfe=rfe, X=X, Key=Key)

        Feat_Importance_Frame.append(summary)

    df_Feature_Gain_Importance = pd.concat(Feat_Importance_Frame, axis=0)
    return df_Feature_Gain_Importance
```

In [84]:
```python
df_Feature_Gain_Importance = Run_Wrapper_Method(Keys=Keys)
```

```
Processing Keys of Wrapper Method: 100%|██████████| 574/574 [34:32<00:00,  3.61s/Key]
```

In [86]:
```python
df_Feature_Gain_Importance = df_Feature_Gain_Importance[df_Feature_Gain_Importance['selected']==True]
df_Feature_Gain_Importance
```

Out[86]:

	Key	feature	selected	rfe_ranking	xgb_gain_importance
0	(10, 1)	ER_t_1	True	1	0.325981
12	(10, 1)	ER_t_1_rollingmean	True	1	0.317084
1	(10, 1)	ER_t_2	True	1	0.125376
40	(10, 1)	week	True	1	0.036234
42	(10, 1)	dayofyear_cos	True	1	0.034948
...	...	...	...	...	...
35	(59, 60)	ER_t_12_rollingstd	True	1	0.042151
24	(59, 60)	ER_t_7_rollingmean	True	1	0.042112
48	(59, 60)	week_cos	True	1	0.041564
29	(59, 60)	ER_t_9_rollingstd	True	1	0.039477
13	(59, 60)	ER_t_1_rollingstd	True	1	0.039340

5740 rows × 5 columns

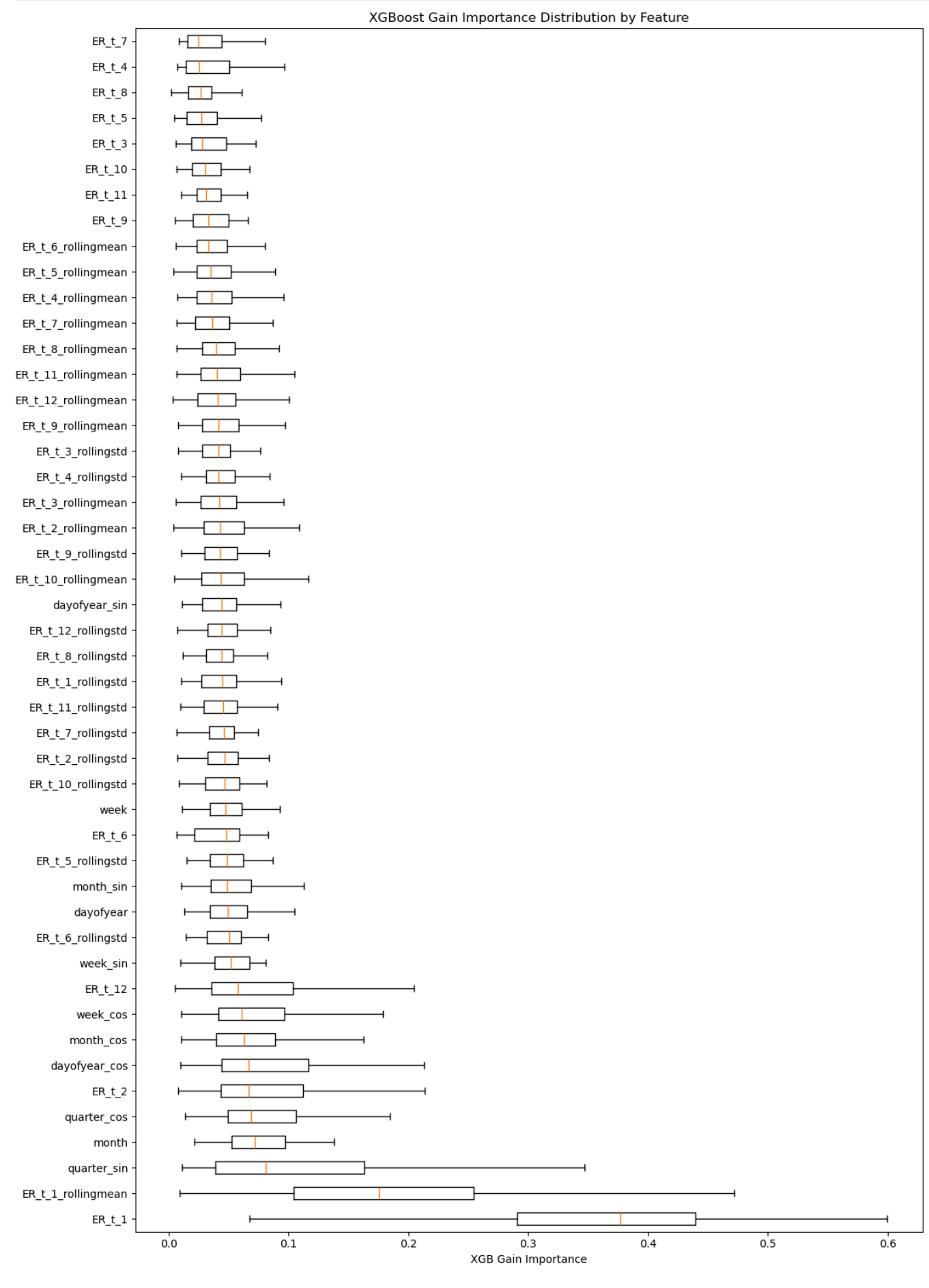
In [93]:
```python
#Sorting Gain Importance based on median (middle of boxplot)
order = (
    df_Feature_Gain_Importance
    .groupby('feature')['xgb_gain_importance']
    .median()
    .sort_values(ascending=False)
    .index
)

#Creating a list of with the sorted distribution
data = [
    df_Feature_Gain_Importance.loc[
        df_Feature_Gain_Importance['feature'] == feature,
        'xgb_gain_importance'
    ]
    for feature in order
]

#ploting the data
plt.figure(figsize=(12, 0.35 * len(order)))
plt.boxplot(data, vert=False, showfliers=False)

plt.yticks(
    ticks=range(1, len(order) + 1),
    labels=order
)

plt.xlabel('XGB Gain Importance')
plt.title('XGBoost Gain Importance Distribution by Feature')
plt.tight_layout()
plt.show()
```



Wrapper Methods -- End -- Outcomes

In [ ]:
```python
'''
Based on the Wrapper Method boxplot lag 1 has generally the highest gain importance, followed by the exogenous variables and
the 2nd and 12th lag along with the rolling mean of lag 1

'''
```

Ensemble Methods -- Start -- Testing Features from Filter and Wrapper Methods with Ensemble Methods

In [94]:
```python
lags_and_indicators
```

Out[94]:
```
['ER_t_1',
 'ER_t_2',
 'ER_t_3',
 'ER_t_4',
 'ER_t_5',
 'ER_t_6',
 'ER_t_7',
 'ER_t_8',
 'ER_t_9',
 'ER_t_10',
 'ER_t_11',
 'ER_t_12',
 'ER_t_1_rollingmean',
 'ER_t_1_rollingstd',
 'ER_t_2_rollingmean',
 'ER_t_2_rollingstd',
 'ER_t_3_rollingmean',
 'ER_t_3_rollingstd',
 'ER_t_4_rollingmean',
 'ER_t_4_rollingstd',
 'ER_t_5_rollingmean',
 'ER_t_5_rollingstd',
 'ER_t_6_rollingmean',
 'ER_t_6_rollingstd',
 'ER_t_7_rollingmean',
 'ER_t_7_rollingstd',
 'ER_t_8_rollingmean',
 'ER_t_8_rollingstd',
 'ER_t_9_rollingmean',
 'ER_t_9_rollingstd',
 'ER_t_10_rollingmean',
 'ER_t_10_rollingstd',
 'ER_t_11_rollingmean',
 'ER_t_11_rollingstd',
 'ER_t_12_rollingmean',
 'ER_t_12_rollingstd']
```

In [98]:
```python
reduced_lags = ['ER_t_1',
 'ER_t_2',
 'ER_t_3', 'ER_t_10',
 'ER_t_11',
 'ER_t_12']

reduced_indicators = [
     'ER_t_1_rollingmean',
     'ER_t_2_rollingmean',
     'ER_t_3_rollingmean',
     'ER_t_4_rollingmean',
     'ER_t_10_rollingmean',
     'ER_t_11_rollingmean',
     'ER_t_12_rollingmean',
]
```

In [ ]:
```python
def Initialize_Models():
    #XG Boost Model with Conservative hyper params:
    #   -n_estimators => build 200 trees
    #   -max_depth => model only 4 levels of feature interaction
    #   -learing rate => learn relatively slow
    xgb = XGBRegressor(
    n_estimators=200,
    max_depth=4,
    learning_rate=0.05,
    subsample=0.8,
    colsample_bytree=0.8,
    random_state=42
    )

    return xgb

def Fit_Model(xgb, X, Y):
    xgb.fit(X,Y)
    booster = xgb.get_booster()
    gain = booster.get_score(importance_type='gain')
    return gain

def Get_Sets(Key):
    train_reduced = train[train['Key']==Key][reduced_lags + reduced_indicators + exog_variables + target]
    X = train_reduced[reduced_lags + reduced_indicators + exog_variables]
    Y = train_reduced[target]

    return X, Y

def Get_Gain_Importance(gain, X, Key):



    xgb_importance_full = np.zeros(X.shape[1])
    gain_series = pd.Series(gain, dtype=float)


    if gain_series.index.str.startswith("f").all():
        gain_series.index = [
            X.columns[int(f[1:])] for f in gain_series.index
        ]


    gain_series = gain_series.reindex(X.columns).fillna(0)

    xgb_importance_full[:] = gain_series.values

    summary = pd.DataFrame({
        "Key": Key,
        "feature": X.columns,
        "selected": xgb_importance_full > 0,
        "xgb_gain_importance": xgb_importance_full
    }).sort_values(
        ["selected", "xgb_gain_importance"],
        ascending=[False, False]
    )

    # Optional ranking (gain-based)
    summary["gain_rank"] = summary["xgb_gain_importance"].rank(
        ascending=False,
        method="dense"
    )

    return summary



def Run_Ensemble_Method(Keys=Keys):

    Feat_Importance_Frame = []
    for Key in tqdm(Keys, desc='Processing Keys for Ensemble Method', unit='Key'):
        X, Y = Get_Sets(Key=Key)

        xgb = Initialize_Models()
        gain = Fit_Model(xgb=xgb, X=X, Y=Y)
        summary = Get_Gain_Importance(gain=gain, X=X, Key=Key)

        Feat_Importance_Frame.append(summary)

    df_Feature_Gain_Importance = pd.concat(Feat_Importance_Frame, axis=0)
    return df_Feature_Gain_Importance
```

In [101...
```python
df_Feature_Gain_Importance_Ensemble = Run_Ensemble_Method()
```

```
Processing Keys for Ensemble Method: 100%|██████████| 574/574 [01:15<00:00,  7.61Key/s]
```

In [102...
```python
df_Feature_Gain_Importance_Ensemble = df_Feature_Gain_Importance_Ensemble[df_Feature_Gain_Importance_Ensemble['selected']==True]
df_Feature_Gain_Importance_Ensemble
```

Out[102...

	Key	feature	selected	xgb_gain_importance	gain_rank
0	(10, 1)	ER_t_1	True	10.073613	1.0
6	(10, 1)	ER_t_1_rollingmean	True	6.135417	2.0
7	(10, 1)	ER_t_2_rollingmean	True	3.747248	3.0
1	(10, 1)	ER_t_2	True	2.886399	4.0
15	(10, 1)	month_sin	True	1.218775	5.0
...	...	...	...	...	...
12	(59, 60)	ER_t_12_rollingmean	True	3.150693	17.0
3	(59, 60)	ER_t_10	True	2.988357	18.0
4	(59, 60)	ER_t_11	True	2.865183	19.0
5	(59, 60)	ER_t_12	True	2.789154	20.0
17	(59, 60)	quarter_sin	True	1.930248	21.0

12046 rows × 5 columns

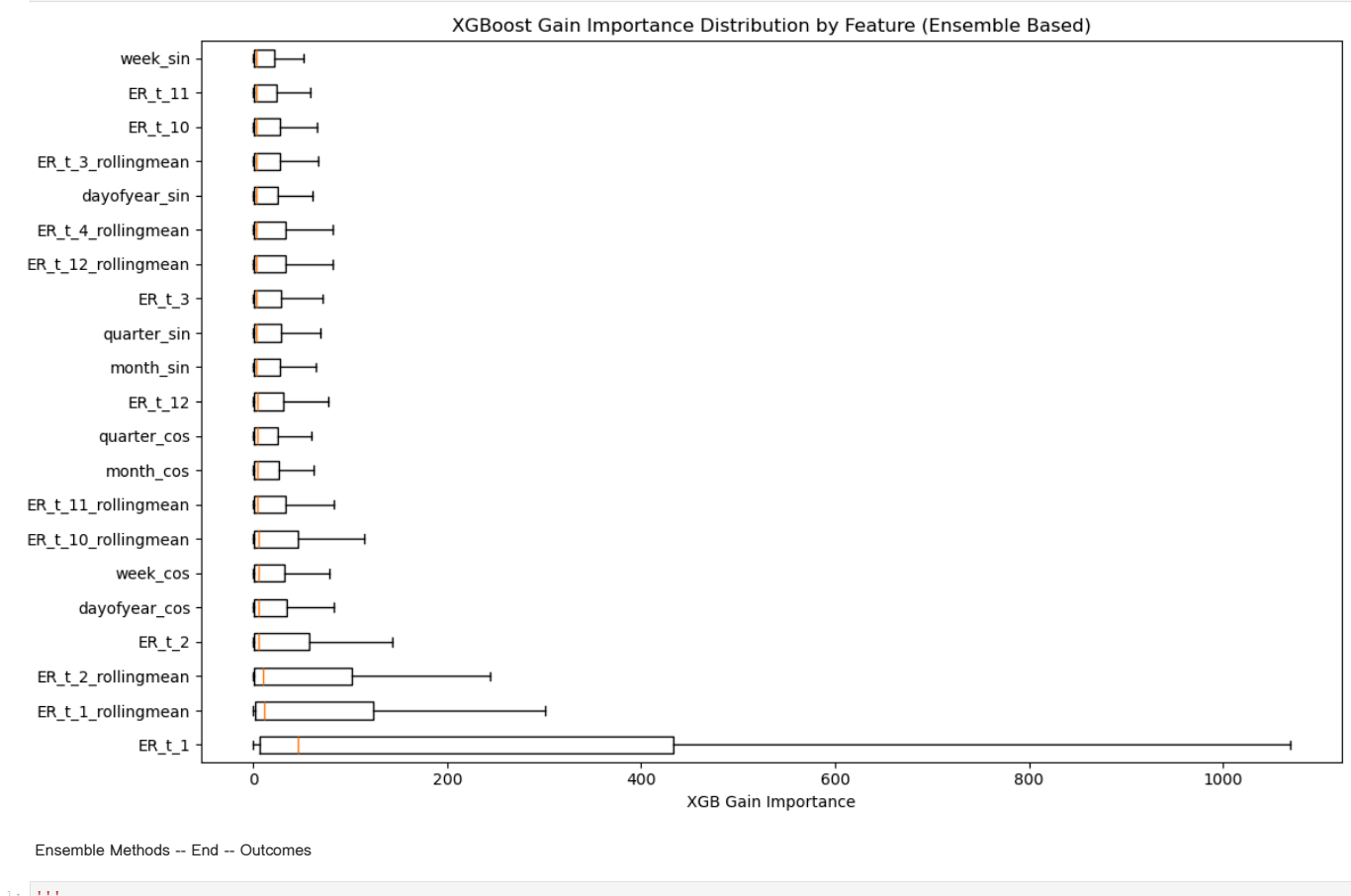
In [103...
```python
#Sorting Gain Importance based on median (middle of boxplot)
order = (
    df_Feature_Gain_Importance_Ensemble
    .groupby('feature')['xgb_gain_importance']
    .median()
    .sort_values(ascending=False)
    .index
)

#Creating a list of with the sorted distribution
data = [
    df_Feature_Gain_Importance_Ensemble.loc[
        df_Feature_Gain_Importance_Ensemble['feature'] == feature,
        'xgb_gain_importance'
    ]
    for feature in order
]

#ploting the data
plt.figure(figsize=(12, 0.35 * len(order)))
plt.boxplot(data, vert=False, showfliers=False)

plt.yticks(
    ticks=range(1, len(order) + 1),
    labels=order
)

plt.xlabel('XGB Gain Importance')
plt.title('XGBoost Gain Importance Distribution by Feature (Ensemble Based)')
plt.tight_layout()
plt.show()
```



Ensemble Methods -- End -- Outcomes

In [ ]:
```python
'''
Based on the Ensemble Results it would seem that Lag 1 is the most dominant feature, followed by the rolling means of lag 1 and 2.
```

```
Furthermore, Lag 12 seems to have relatively high gain when compared to Lag 3 and Lag 10. However, we are looking at distribution of
gain, therefore it might be pertitent to include lags 1, 2, 3, 10, 11, 12 along with rolling mean indicators for 1, 2, 10, 11.
'''
```